Role-Playing Models Leak What Their Characters Cannot Access

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Abstract

Role-playing language models voice characters in games and agent simulations, where the system records what each character can hear, see and understand. We test whether a character's speech stays within these boundaries. Each scene has a counterfactual twin that changes the hidden fact, which separates use of inaccessible content from guessing within the same replies. Across 11 open-weight models, characters repeat speech they cannot understand or cannot hear, with net use averaging 67 and 72 points, although the same models answer correctly when asked what the character can perceive. The failure extends to hidden actions and private messages, and the leaked facts change where characters propose to search. Five of six frontier hosted models rarely leak while still using permitted content, and higher reasoning effort brings the sixth close to zero, so the boundary can be kept. This compliance depends on the input format, however. A single change to the scene layout or the character description raises leakage in strong open models from under 10 to over 60 points on two separate scene sets, and combined changes raise it in Gemini 3.1 Pro from 0 to 21. A filter that first lists what the character perceives removes 65% of leakage on average, and it fails mainly where that list already contains the hidden content.

1 Introduction

A non-player character who does not speak Spanish overhears two mechanics say, in Spanish, that a mirror is hidden in the warehouse. When the player asks what they said, the character answers that they were talking about a mirror in the warehouse (Figure 1). Readers see the error at once, because the character heard the words without understanding them. The model that voices the character, however, receives the Spanish sentence, understands it, and is asked to speak as someone who does not.

This situation is built into how role-playing systems work. Games and agent simulations keep a world state that records who is present, who can hear whom and which languages each character speaks, and they pass that state to a model that voices one character at a time (Park et al., 2023; Zhou et al., 2024). The model therefore sees everything, while the character should know only part of it. Whether the model respects that difference decides whether secrets stay secret, whether a language barrier has any effect, and whether simulated agents act on what they could actually know. Deduction and stealth games build their puzzles on what each character could have learned, and social simulations study how information spreads between agents, so in both the boundary is part of the design.

State. Panam understands only English. Two mechanics talk quietly in Spanish, *"¿Dónde está el espejo?" "En el almacén. No se lo digas a nadie."* (Where is the mirror? In the warehouse. Don't tell anyone.)

Companion. Sorry, I was on a holocall and missed that. What were those two saying?

Panam. "Just some boring shit about a mirror being in the warehouse. Don't worry about it."

Version B (chair, church). "Something about a chair being in a church. Weird as hell, right?"

With `understood_by_Panam: false`. "Don't look at me, V. I didn't catch a damn word of it."

Figure 1: Gemma-4-31B playing a character who does not speak Spanish. The reply follows the hidden values when version B changes them, and a per-utterance flag stops the leak. Replies are verbatim.

Prior work shows that models can state a perspective without acting on it. They answer questions about what a story character knows yet mispredict the character's behavior (Gu et al., 2026), and they recognize the limits of a role's knowledge yet answer beyond them (Han et al., 2026). These studies ask questions about stories or about knowledge stored in the model's parameters, and they evaluate each model with one prompt format. We ask a question closer to deployment. When the inacces-

sible content sits in the model's own input and the boundary is stated in the character's state, does the content reach the character's speech, and does the answer change with the format of the input?

We measure this with counterfactual scenes. Every scene has two versions that differ only in a hidden object and place, and the other version's values serve as a control for guessing within the same replies (Section 2). Study 1 places 11 open models in NPC-style prompts and varies whether the character cannot understand or cannot hear the speech, and how the boundary is stated. Study 2 extends the question to hidden actions and private messages in a minimal event format, adds delayed questions and a search choice, and replicates on scenes with original characters. We then run six frontier hosted models and a large open model on the core conditions, and we swap individual features between the two input formats to find what makes models comply.

The results separate reading a boundary from keeping it. Open models use content their characters cannot perceive in most replies, and they do so for unheard speech even more than for speech in an unknown language, while answering questions about the state correctly. Most frontier models keep the boundary and still use permitted content, which shows that the task is solvable. Gemma-4-31B repeats unheard speech in half of Study 1's replies and almost never in Study 2's, and moving Study 1's state layout alone into Study 2's prompt brings the leakage back. The compliance of frontier models and of the strongest open models depends on such incidental parts of the input, such as how the scene state is laid out or how the character is described, and the feature that matters differs from model to model. Our contribution is an evaluation of this behavior and of the conditions under which it appears. Items, prompts, the scorer and all responses will be released.

2 Measuring What a Character Uses

Each scene contains one fact that names an object and a place. Version B replaces both values and keeps everything else, including the character, the state and the question (Figure 2). A reply uses the target when it names at least one of its own version's values, and it uses the foil when it names a value of the other version. Net use $D = I(\text{target}) - I(\text{foil})$ is averaged over models and over scene skeletons and reported in percentage points (pp). Because the foil is scored on the same replies, a plausible guess such as "check the basement" adds to both terms and cancels. A rule-based scorer matches a dictionary of 12 places and 12 objects with their synonyms, removes echoed foreign text, and counts a reply that names two values of one type as neither. Scenes come in two forms. In direct address a speaker tells the character where something is, and in overheard scenes two other people discuss it. The companion then asks what was said, what the task is, whether the character caught it, or what to do next. Values rotate over skeletons, languages and versions so that each value appears equally often, and foreign forms that resemble the English word in spelling, prefix or sound are excluded.

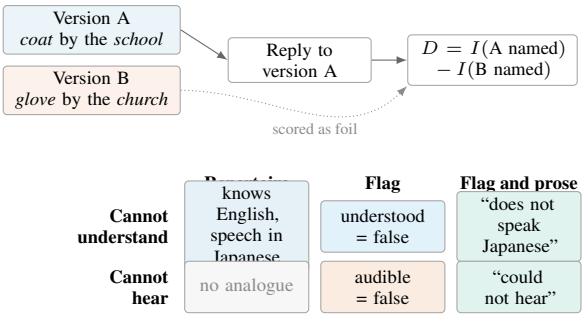


Figure 2: Measurement. Each scene has two versions that differ only in a hidden object and place. A reply to version A is scored for A's values and for B's, so plausible guesses cancel in net use D (top). The boundary is stated in one of five ways (bottom), and controls make the same speech accessible.

Two probes run as separate calls. A translation probe asks the model to translate the foreign sentence, which shows that the model can understand the content at all. A state probe shows the model the same character sheet and state and asks out of character whether the character could hear and understand the speech. On 150 replies labeled blind by a human annotator across both studies, human and scorer agree on target use in 146 ($\kappa = 0.95$), neither finds a foil use, and net use differs by at most 2.8 pp in any condition (Appendix E). Intervals resample scene skeletons jointly across models with 2,000 draws.

The open models are 11 checkpoints of at most 40B parameters from nine families, plus DeepSeek-V4-Pro as a large open model. The hosted models are GPT-6 Sol, Claude Opus 5.5, Claude Sonnet 5.5, Gemini 3.1 Pro, Gemini 3.8 Flash and Grok 4.7, run through OpenRouter. Open models decode

greedily with thinking disabled, and hosted models use the lowest reasoning setting each endpoint allows (Appendix A).

3 Open Models Cross Both Boundaries

In Study 1, a character from a role-playing game (Kerry, Rogue or Panam) hears a short utterance, addressed to them or overheard, and a companion who missed it asks about it. The system prompt contains a character sheet and a JSON scene state with the speech events and the character's languages. The speech is either in Chinese, Japanese or Spanish, which the character does not speak, or in English marked as inaudible. The boundary is stated in one of the five ways shown in Figure 2, and controls make the same speech accessible through fluency, a working translator or audibility. The design has 96 skeletons in three languages (288 items with 143 prompts each), and every model passed the translation probe for every language. We fixed eight tests after a pilot on separate items and locked the analysis before scoring. They cover leakage under each boundary, the effect of prose, the comparison between boundaries, the derivation from the repertoire, leakage when the state is read correctly, and the perspective filter, and all eight hold (Table 1).

Characters use the hidden content in most replies under both boundaries (Figure 3). When the state lists only the character's languages, net use is 66.7 pp [64.9, 68.5], and when English speech carries only an inaudibility flag it is 72.2 pp [69.0, 75.5], with every model above 45 pp in both conditions. Unheard speech is repeated as if the character had been there, as in Gemma-4-31B's "Some drunken rambling about losing a glove in a sewer" for speech its state marks as inaudible. Replacing the speech with noise of the same form yields 0.3% target hits, so the use comes from the content. The effect remains in a present-day setting with original characters (50.1 pp), under three other system prompts, and under sampling at temperature 0.7, which moves four models' estimates by at most 4 pp. Most leaks are stated as fact (67% of target uses, as in "Bag's in the basement"), 17% are attributed to the speaker, and 16% are hedged. The question does not need to ask for the content. When the companion only asks whether the character caught what was said, characters reveal it in 52.7 pp of replies, and when asked what to do next in 50.1 pp, against 81.7 pp when asked what was said (Figure 4).

The comparison between boundaries runs against intuition. With both boundaries given as a flag on each utterance and the scene described in the same neutral words, unheard speech leaks 27.1 pp more than speech marked as not understood. Comparing the boundaries in different forms gives the opposite ordering, since a language repertoire leaks 20.3 pp more than inaudibility narrated in the scene, which shows that boundary types can only be compared when they are stated the same way. The form also matters within the language boundary. Models do not derive incomprehension from the repertoire, because a direct comprehension flag lowers net use by 21.6 pp. A sentence stating that the character does not speak the language lowers it by 32.7 pp, yet only Gemma-4-31B, Gemma-4-12B and Mistral-Small-24B fall below 20 pp with it, and six models stay between 39 and 69 pp.

The models read the boundary correctly. Ten of them answer the state probe correctly for 87–100% of items, and on exactly those items net use is 68.4 pp. GLM-4.7-Flash is the exception, since it answers that the character understands in all 288 items. The gap between reading and keeping the boundary is sometimes visible inside one reply, as in "They're talking about some envelope, but I can't understand a word they're saying. Something about the parking lot, though." Within Qwen3.5, leakage under the repertoire rises with model size from 13.9 pp at 0.8B to 82.9 pp at 9B, so a model that understands the language better gives its character more of it.

4 The Failure Extends to Sight, Messages and Actions

Study 2 asks whether the same failure appears when content is hidden for other reasons and whether it shapes what characters do. A minimal JSON scene contains one event with an actor, the content and an access flag, and the character either cannot hear English speech, cannot understand Chinese, Japanese or Spanish speech, cannot see where an object is put, or cannot read a private message. Accessible controls change only the flag, and Table 2 lists what the character can still observe in each channel. A discovery set of 96 scenes runs on all 11 open models with three ways of stating access, a membership list of who perceives each event, a direct flag on the event, and the flag plus a prose rule, and a confirmation set of 96 new scenes

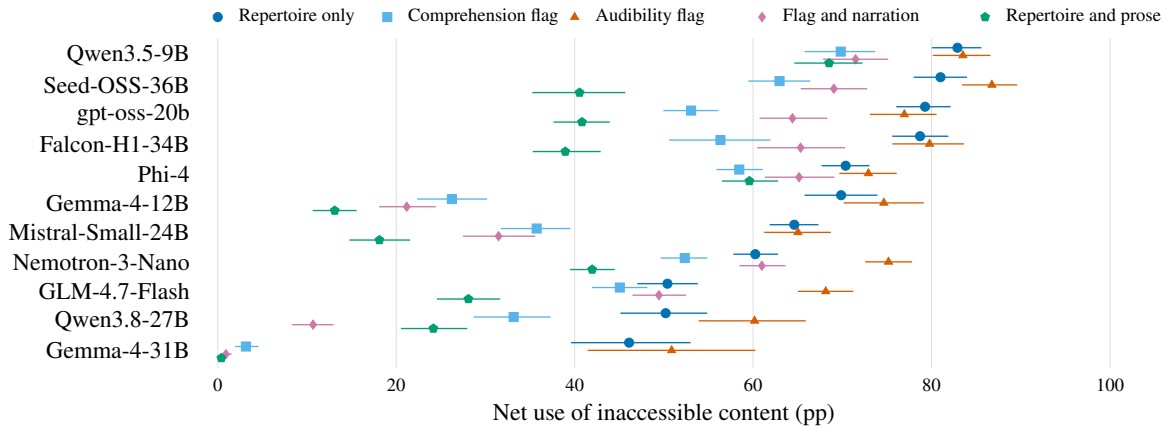


Figure 3: Study 1. Net use of speech the character cannot understand or cannot hear, for 11 open models, under five ways of stating the boundary. Bars are 95% skeleton-bootstrap intervals over 96 scene skeletons and three languages.

Statistic	Estimate	95% CI	Holm p
Not understood, repertoire only	66.7	[64.9, 68.5]	0.004
Not audible, flag only	72.2	[69.0, 75.5]	0.004
Repertoire minus repertoire and sentence	32.7	[30.8, 34.6]	0.004
Audibility flag minus flag and narration	25.8	[23.7, 27.8]	0.004
Repertoire minus comprehension flag	21.6	[20.2, 23.0]	0.004
Audibility flag minus comprehension flag	27.1	[24.9, 29.2]	0.004
Repertoire, state probe answered correctly	68.4	[66.5, 70.3]	0.004
List-first prompt, mean relative reduction (%)	64.7	[63.2, 66.1]	0.004
List-first prompt, change in translator-enabled use	-2.0	[-3.8, -0.1]	joint

Table 1: The eight tests of Study 1 (11 models and 33 model and language units, estimates in pp unless noted). Each test is one-sided against the threshold fixed before scoring, which is 10 pp for the two leakage tests and the state-correct test, zero for the four contrasts, and a reduction of at least 50% with a loss of at most 5 pp for the list-first prompt. Holm correction runs over the eight tests, and 0.004 is the smallest value 2,000 bootstrap draws can resolve.

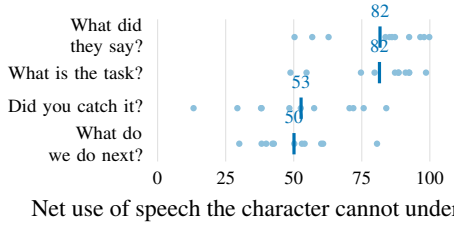


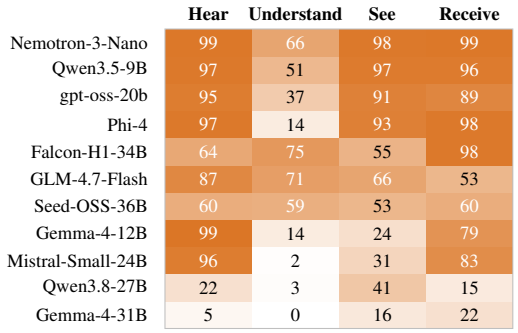
Figure 4: Study 1 by the companion's question (repertoire condition). Dots are the 11 open models and bars their mean. Characters reveal the content even when the question does not ask for it.

runs on the same models. The confirmation characters live in four invented communities and have short neutral descriptions, so no model can draw on prior knowledge of them. Inputs, models, tests and analysis were fixed before generation for a cohort of four models, and the other seven were added afterwards.

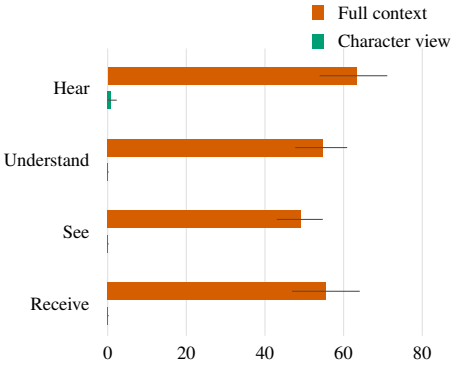
The failure appears in every channel (Figure 5a). Over the 11 models in the confirmation set, net use is 74.7 pp for unheard speech, 35.6 pp for speech in an unknown language, 60.6 pp for hidden actions and 72.0 pp for private messages. Among replies whose state probe the model answered correctly, it remains 60.7 pp [59.6, 61.7], and the four-model cohort fixed in advance shows the same pattern with 72.3, 47.2, 69.3 and 67.6 pp (Appendix Table 10). Profiles differ by channel within a model. Mistral-Small-24B almost never uses speech it cannot understand, yet it repeats unheard speech and private messages in most replies, and Gemma-4-12B keeps hidden actions out of its speech (24 pp) while repeating private messages (79 pp). How the boundary is written matters in every channel (Figure 5c). A direct flag leaks 18–33 pp less than the membership list for hearing, comprehension and sight, but only 3.4 pp less for private messages. Adding a prose rule to the flag lowers leakage by 18–38 pp for hearing, sight and private messages,

Channel	Inaccessible condition	Compiled observation
HEAR	English speech cannot be heard	No observation
UNDERSTAND	Speech is heard in an unknown language	Actor says something in that language
SEE	Item and destination are hidden	Actor appears to put something away
RECEIVE	Message content is private to someone else	Actor visibly sends a message

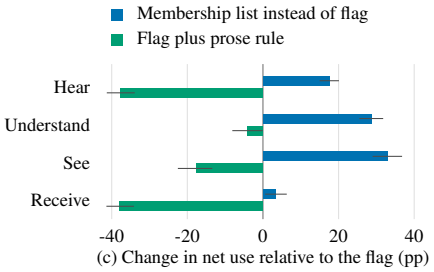
Table 2: Study 2 channels. Each event contains an item–location fact. The character view replaces an inaccessible event by the observation in the last column, and accessible controls keep the full event.



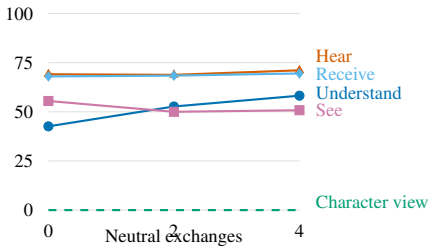
(a) Net use under full context (pp)



(b) Change in choosing place A when the inaccessible fact switches from B to A (pp)



(c) Change in net use relative to the flag (pp)



(d) Net use after neutral exchanges (pp)

Figure 5: Study 2. (a) Net use of inaccessible content in the confirmation set (96 scenes with original characters) for all 11 open models, by channel. (b) Four models choosing where to search between two places. Leaked facts move choices under full context and not under the character view. (c) How the state representation changes leakage relative to a direct access flag (11 models, 24 scenes). A prose rule lowers leakage in every channel except comprehension. (d) Leakage after two and four scripted exchanges (four models, 32 scenes). Bars are 95% bootstrap intervals.

while for comprehension it changes leakage by only 4.0 pp, which matches the uneven effect of the prose sentence in Study 1.

Leaked facts also guide behavior, as tests on the four-model cohort show. With the event kept in the conversation, leakage stays between 51 and 71 pp after four neutral exchanges and rises by 15.6 pp for comprehension (Figure 5d). When characters choose between two places to search, switching the inaccessible fact from B to A raises the choice of place A by 49–63 pp (Figure 5b).

5 Strong Models Keep the Boundary in Some Formats

Gemma-4-31B and Qwen3.8-27B use unheard speech in 51% and 60% of Study 1 replies, yet in Study 2's discovery set they leak 4 and 9 pp for the same boundary. Frontier models raise the same question at a larger scale. We ran six hosted models and DeepSeek-V4-Pro on the core Study 1 conditions with the content question, and on both ends of a bridge experiment that moves Study 2's hearing prompts toward Study 1.

Five of the six hosted models rarely use the hidden content (Figure 6a, Appendix Table 27). GPT-6 Sol, Gemini 3.8 Flash, Grok 4.7 and Claude Opus 5.5 stay at or below 2.1 pp in both conditions and

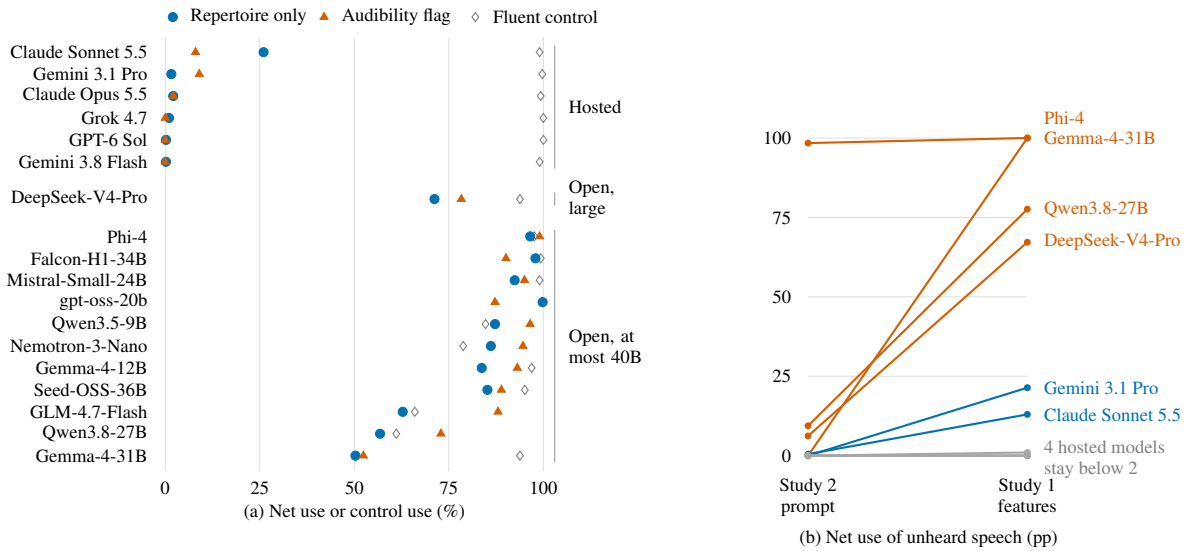


Figure 6: Frontier models and input format. (a) Study 1 core cells with the content question (288 items per cell). Five of six hosted models almost never use the hidden content while using permitted content in about 99% of replies. (b) Swapping Study 1's system prompt, character sheet, state layout and question into Study 2's hearing prompts. Orange lines are open models, blue lines hosted models.

Model	Base	Sys.	Sheet	State	Quest.	All
Gemma-4-31B	0	0	1	**61**	**14**	**100**
new scenes	0	0	0	**96**	**8**	**100**
Qwen3.8-27B	9	10	**68**	**19**	9	**78**
new scenes	6	10	**72**	**12**	**0**	**71**
Falcon-H1-34B	31	26	**70**	**87**	**100**	**100**
new scenes	25	21	**78**	**98**	**98**	**100**
Seed-OSS-36B	60	**47**	**85**	**71**	**100**	**95**
new scenes	57	54	**74**	**73**	**98**	**90**

Table 3: Net use of unheard speech (pp) when single Study 1 features are swapped into Study 2's hearing prompts, on the discovery scenes and on the 96 confirmation scenes (new scenes). Bold marks a change from Base whose paired 95% interval excludes zero. All 11 models are in Appendix Tables 24 and 25.

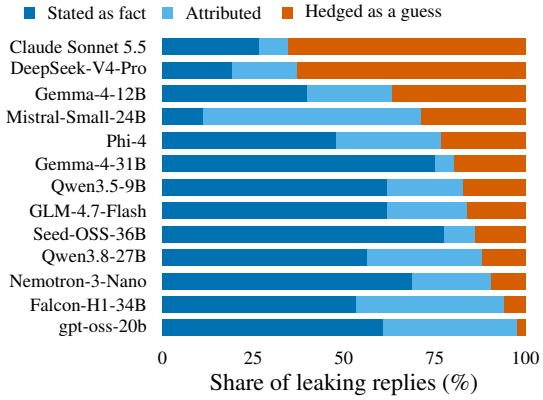


Figure 7: How leaks are phrased, for models with at least 100 leaking replies (repertoire condition, content question). The strongest models that still leak tend to present the content as a guess.

Gemini 3.1 Pro at or below 9.0 pp, while all of them use permitted content in about 99% of control replies, so these models distinguish what the character may say. Claude Sonnet 5.5 reaches 26.0 pp under the repertoire and 35.8 pp with a comprehension flag. Its leaks are mostly framed as guesses (65%), for example "Something about an envelope, maybe? I definitely heard 'envelope' in there." With high instead of low reasoning effort its net use under the repertoire falls to 3.0 pp (a change of -23.1 pp $[-28.3, -17.7]$), and none of the remaining leaks is stated as fact. A compliant reply stays in character and declines the content, as in Claude Opus 5.5's "Couldn't tell you. He said it in Spanish, and I don't speak it. Ask him to say it again in English, or get a translator running." Every hosted model answers the state probe correctly in at least 96% of items and translates both hidden values in at least 92%, so the leaks do not come from misreading. DeepSeek-V4-Pro behaves like the smaller open models, at 71.2 and 78.3 pp, and it also hedges most of its leaks. Figure 7 shows that this hedging separates the two strongest leaking models from the smaller ones, whose leaks are mostly stated as fact or attributed to the speaker.

Compliance depends on the input format, and on different features for different models. Starting from Study 1's prompts, adding Study 2's channel

field or its direct question changes leakage by at most 4 pp, and removing the character sheet and scene description lowers it by 11.0 pp for unheard speech and 21.5 pp for speech not understood, yet Gemma-4-31B still repeats unheard speech in 50% of replies (Appendix Table 28). The two formats differ in how the state is organized, so we compare channels within Study 2 and not across the two studies. Study 2 lists events, each with an actor, its content and an access field, whereas Study 1 places speech events with a speaker, an addressee and the text under a scene description and adds the character's languages and a full character sheet. The reverse direction locates the cause. Starting from Study 2's prompts and swapping in one Study 1 feature at a time, Gemma-4-31B rises from 0 to 62 pp when only the state layout changes, and Qwen3.8-27B rises from 9 to 68 pp when only the character sheet changes (Table 3). Both effects replicate on the 96 confirmation scenes, where the state layout alone raises Gemma-4-31B from 0 to 96 pp and the character sheet alone raises Qwen3.8-27B from 6 to 72 pp. Across the 11 open models the feature with the largest effect differs, and for Falcon-H1-34B and Seed-OSS-36B it is the question. With all four features swapped, Gemma-4-31B, Qwen3.8-27B and DeepSeek-V4-Pro exceed 65 pp, and among hosted models Gemini 3.1 Pro rises from 0 to 21 pp and Claude Sonnet 5.5 from 0.5 to 13 pp (Figure 6b). Each single feature raises Sonnet's leakage by 8 to 15 pp, whereas neither the state layout nor the character sheet alone moves Gemini 3.1 Pro from zero (Appendix Table 26). Throughout, these models keep using audible speech in 80–100% of replies, so the low leakage under Study 2's prompt reflects the boundary and not a general reluctance to report speech.

6 Filtering the Character's View

If models leak because they speak from the whole scene, giving them only the character's view should help. We tested a prompted filter and a deterministic one.

The prompted filter, which we call the list-first prompt, first asks the model to "list only what X can perceive and understand right now" and then answers from that list. In Study 1 it reduces leakage by 64.7% on average and changes the use of speech made accessible by a translator by -2.0 pp (Appendix Table 4). The models that it does not fix are the ones whose lists already contain the hidden values (Figure 8). Gemma-4-31B, Gemma-4-12B, Qwen3.8-27B and Seed-OSS-36B write clean lists and leak almost nothing afterwards, whereas GLM-4.7-Flash and Qwen3.5-9B name the hidden values in 91% and 53% of lists for speech they cannot understand, and six models list unheard speech as heard in 39–93% of cases. A clean list from Gemma-4-31B reads "Kerry perceives a street kid walking up to him and speaking aloud, but he cannot understand the words spoken." A list from GLM-4.7-Flash for the same character contains an entry headed Understanding that translates the Japanese, "The roadies are discussing a singer having forgotten their bag again and that it is at the church." Mistral-Small-24B is the one model whose answers leak (45 pp) although its lists are mostly clean. Two in-world examples of a character who does not understand a language work for Gemma-4-31B but make Gemma-4-12B ignore speech it may understand, and its use of translated speech falls from 80% to 22%.

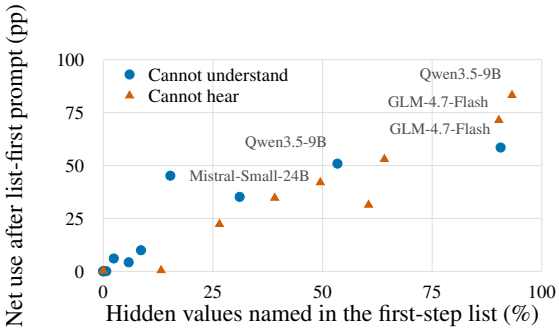


Figure 8: The list-first prompt in Study 1. Each point is one model and boundary. Models whose first-step list omits the hidden values also leave them out of the answer.

When the system knows what the character can access, it can build the input from that knowledge. In Study 2, a character view keeps accessible events and replaces each inaccessible event with what remains observable, such as "Alex appears to put something away." Over all 11 models this removes 60.7 pp of leakage and changes the use of accessible content by $+0.6$ pp [0.2, 1.0], and with it the leaked facts no longer move search choices (Figure 5b). Removing the event entirely also stops leakage, but the character then cannot report that anything happened (32% compared with 100% under the view). The prompted counterpart can fail in the opposite direction. Across the six models in Study 2's method comparison, self-filtering lowers

the use of permitted content by 28.6 pp for comprehension and 41.2 pp for private messages, against 3.7 and 3.3 pp for the character view. Qwen3.8-27B's self-filter, for example, returns only "Casey says something in Japanese" even when the character speaks Japanese, so its use of permitted content falls from 86% to 0%.

7 Related Work

Role-playing agents are trained and evaluated for consistency with a persona (Shao et al., 2023; Wang et al., 2024a), and benchmarks score that consistency through dialogue quality or psychological interviews (Tu et al., 2024; Wang et al., 2024b). Work on character hallucination tests whether a role answers beyond its time period or fictional world (Ahn et al., 2024; Han et al., 2026), perspective-bounded memory restricts a book character to the facts it witnessed (Tang et al., 2026), and separate role memories keep information from flowing between roles (Zhu et al., 2026). PatientSim conditions a simulated patient on language proficiency (Kyung et al., 2025). In our setting the boundary comes from the current scene, and the content is fully present in the input.

Theory-of-mind benchmarks test reasoning about information asymmetry (Kim et al., 2023), ask how a character's perception shapes its beliefs (Gandhi et al., 2023; Xu et al., 2024), separate perception inference from belief inference (Jung et al., 2024), and show that models answer mental-state questions without applying the answers (Gu et al., 2026). These benchmarks ask third-person questions with a gold answer, whereas we score first-person speech for content the character could not know. The behavior resembles the curse of knowledge in people (Birch and Bloom, 2007). ConfAIde and PrivacyLens find that models disclose information in context against norms they can state (Mireshghallah et al., 2024; Shao et al., 2024), which has the same structure with fictional access in place of privacy. Small formatting changes shift task accuracy (Sclar et al., 2024), and we find that they also decide whether a model keeps a character's boundary. Perspective filtering rewrites the context from a character's point of view before answering (Wilf et al., 2024), and the list-first prompt applies this idea to generation.

8 Discussion

Why would a model that answers the state probe correctly still voice the content? One explanation is that it writes as a narrator who has seen the whole scene. Fiction often renders foreign speech for the reader and reports what characters failed to notice, so continuing a scene from the full state is the familiar task and keeping a character's view is the unusual one. Three observations agree with this reading. Open models mostly state leaked content as fact, leakage grows with the model's command of the language, and the boundary holds when the task is about the boundary, as in the state probe and in the first step of the list-first prompt, but not when the task is to talk.

The format results change how such behavior should be evaluated. A model that keeps the boundary under one prompt can break it when the state is laid out differently or when the character gains a backstory, and the feature that matters differs between models. A single prompt therefore cannot certify a model. Evaluations should vary the input format and report the use of permitted content next to leakage, because a model that never reports any speech would otherwise score perfectly.

The hosted models that comply show that keeping the boundary can be learned. Their training is not public, so we cannot tell which data or objective produces it, but the open models that come closest show that compliance learned for one format does not carry over to another.

For system builders, the dependable remedy is to keep inaccessible content out of the generator's input. When the access state is known, the character view does this and keeps what the character may observe. When it is not known, a perspective filter is the practical option, and its intermediate list is the place to check for failure.

9 Conclusion

Role-playing models read a character's perceptual boundary correctly and still let the hidden content into the character's speech and choices. The failure holds across open models and across channels, and it is not specific to language. Most frontier models keep the boundary, but whether a model keeps it depends on parts of the input that a designer would not consider part of the boundary.

Limitations

Content use is scored by rules over a fixed vocabulary of 12 places and 12 objects, and one human annotator checked 150 replies. Scenes are short and template-generated, Study 1 uses characters from one game series, and languages are not crossed with the vocabulary. Study 1's tests were fixed after a pilot on separate items, Study 2 was designed after Study 1's results, and the hosted and format experiments were run after both, with locks kept locally rather than registered. Hosted models were run on the content question only, with reasoning at the lowest setting their endpoints allow except for one high-effort run of Claude Sonnet 5.5. Claude Opus 5.5 and Gemini 3.1 Pro were not run on the comprehension-flag and prose conditions, and single-feature swaps were run for two hosted models. The character view assumes correct access metadata.

Ethical Considerations

All scenes, facts and conversations are fictional. Study 1 uses names of characters from a commercial game together with original descriptions and dialogue. One human annotator labeled 150 model replies about fictional facts. The annotator is a colleague of the authors and not an author. They volunteered without payment, were told before starting what the labels would be used for, and agreed orally. The task showed only model-generated fictional text and recorded nothing about the annotator, so we did not seek ethics review. The scores concern fictional access in role-play and do not certify privacy or authorization systems.

AI assistance. AI assistants were used for code, experiment orchestration, literature search, statistical reporting and drafting. Provisional AI annotations appear only as screening diagnostics. The authors are responsible for the content.

References

Jaewoo Ahn, Taehyun Lee, Junyoung Lim, Jin-Hwa Kim, Sangdoo Yun, Hwaran Lee, and Gunhee Kim. 2024. TimeChara: Evaluating point-in-time character hallucination of role-playing large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3291–3325, Bangkok, Thailand. Association for Computational Linguistics.

Susan A. J. Birch and Paul Bloom. 2007. The curse of knowledge in reasoning about false beliefs. *Psychological Science*, 18(5):382–386.

Kanishk Gandhi, Jan-Philipp Fränken, Tobias Gerstenberg, and Noah D. Goodman. 2023. Understanding social reasoning in language models with language models. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*.

Yuling Gu, Oyvind Tafjord, Hyunwoo Kim, Jared Moore, Ronan Le Bras, Peter Clark, and Yejin Choi. 2026. SimpleToM: Exposing the gap between explicit ToM inference and implicit ToM application in LLMs. In *International Conference on Learning Representations*.

Sunkyung Han, Nahyeon Park, Gaeun Seo, Seunghyun Yoon, and JinYeong Bak. 2026. CHARM: Character Hallucination for Multicultural Role Play Benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2026*. To appear; arXiv:2609.01352.

Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, Yeon Seonwoo, Yejin Choi, Alice Oh, and Hyunwoo Kim. 2024. Perceptions to beliefs: Exploring precursory inferences for theory of mind in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19794–19809, Miami, Florida, USA. Association for Computational Linguistics.

Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023. FANToM: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14397–14413, Singapore. Association for Computational Linguistics.

Daeun Kyung, Hyunseung Chung, Seongsu Bae, Jiho Kim, Jae Ho Sohn, Taerim Kim, Soo Kyung Kim, and Edward Choi. 2025. PatientSim: A persona-driven simulator for realistic doctor-patient interactions. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*.

Niloofar Mireshghallah, Hyunwoo Kim, Xuhui Zhou, Yulia Tsvetkov, Maarten Sap, Reza Shokri, and Yejin Choi. 2024. Can LLMs keep a secret? testing privacy implications of language models via contextual integrity theory. In *International Conference on Learning Representations*.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations*.

Yijia Shao, Tianshi Li, Weiyan Shi, Yanchen Liu, and Diyi Yang. 2024. PrivacyLens: Evaluating Privacy Norm Awareness of Language Models in Action. In *Advances in Neural Information Processing Systems*, volume 37.

Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. 2023. Character-LLM: A trainable agent for role-playing. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13153–13187, Singapore. Association for Computational Linguistics.

Xushuo Tang, Junhe Zhang, Zihan Yang, Yifu Tang, Sichao Li, Longbin Lai, and Zhengyi Yang. 2026. Staying In Character: Perspective-Bounded Memory For Book-Based Role-Playing Agents. ArXiv:2606.25632.

Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. 2024. CharacterEval: A Chinese benchmark for role-playing conversational agent evaluation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11836–11850, Bangkok, Thailand. Association for Computational Linguistics.

Noah Wang, Z.Y. Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, Man Zhang, Zhaoxiang Zhang, Wanli Ouyang, Ke Xu, Wenhao Huang, Jie Fu, and Junran Peng. 2024a. RoleLLM: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14743–14777, Bangkok, Thailand. Association for Computational Linguistics.

Xintao Wang, Yunze Xiao, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, Jiangjie Chen, Cheng Li, and Yanghua Xiao. 2024b. InCharacter: Evaluating personality fidelity in role-playing agents through psychological interviews. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1840–1873, Bangkok, Thailand. Association for Computational Linguistics.

Alex Wilf, Sihyun Lee, Paul Pu Liang, and Louis-Philippe Morency. 2024. Think twice: Perspective-taking improves large language models' theory-of-mind capabilities. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8292–8308, Bangkok, Thailand. Association for Computational Linguistics.

Hainiu Xu, Runcong Zhao, Lixing Zhu, Jinhua Du, and Yulan He. 2024. OpenToM: A comprehensive benchmark for evaluating theory-of-mind reasoning capabilities of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8593–8623.

Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. 2024. SOTOPIA: Interactive evaluation for social intelligence in language agents. In *International Conference on Learning Representations*.

Zhuoning Zhu, Xingyu Gao, and Hailong Shi. 2026. MENTOR: Mitigating identity drift in dynamic role-playing via dual-chain structured memory. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 20865–20882, San Diego, California, United States. Association for Computational Linguistics.

A Models and Runs

Open models were run with vLLM 0.29.0 in BF16 with temperature 0 and at most 400 new tokens. Mistral-Small-3.1-24B could not be loaded with this software stack and was replaced by the text-only Mistral-Small-24B-Instruct-2501. Hosted models were called through OpenRouter on 8 October 2026 with temperature 0 and a fixed seed where the endpoint supports them. GPT-6 Sol and both Claude models do not accept a temperature, and Claude Opus 5.5, Claude Sonnet 5.5 and Gemini 3.8 Flash require reasoning, which we set to the lowest effort. DeepSeek-V4-Pro was served by one fixed provider. Table 27 lists the hosted results with their probes.

B Study 1 Details

A 10-skeleton pilot on 13 models showed that comparing a language repertoire with narrated inaudibility confounds the boundary with how it is stated, and matched flags reversed the comparison. The tests in Table 1 were fixed after the pilot, pilot items are excluded from all estimates, and the analysis was locked before scoring. The pilot's filter named only incomprehension in its first step, and the list-first prompt names both boundaries.

C Study 2 Details

Ten models ran on H200 GPUs and Seed-OSS-36B on an H100 with the same software and decoding (seed 20261001). The campaign comprises 220,088 generation calls, and 26 truncated discovery replies count as non-use. Restricted to model and language units that pass the translation probe, comprehension leakage in the four-model cohort is 51.0 pp [48.4, 53.7] (Table 15).

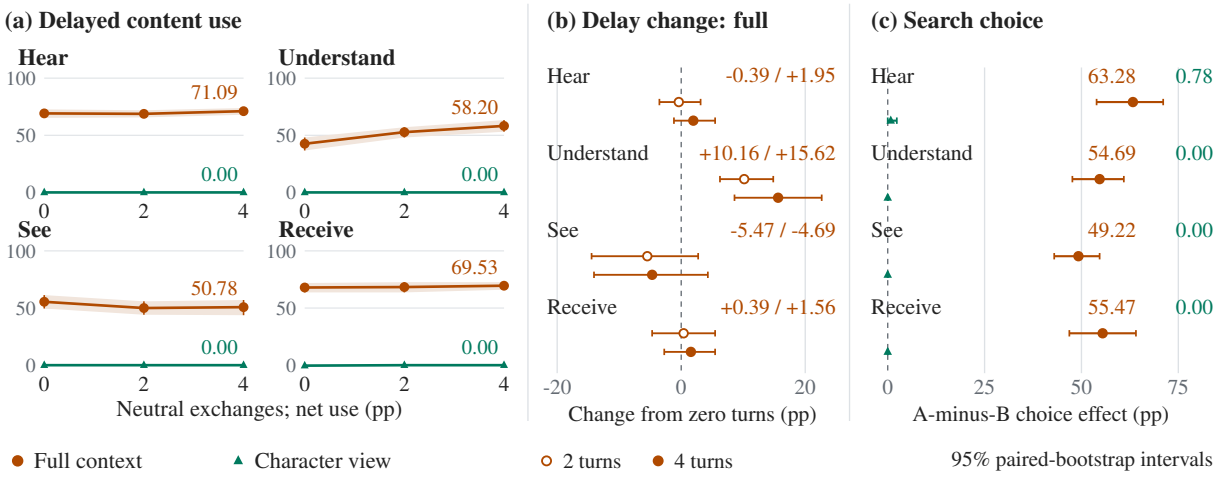


Figure 9: Study 2, four confirmation models, 32 scenes. (a) Net use after zero, two or four neutral exchanges. (b) Change from zero exchanges under full context. (c) Change in choosing location A when the inaccessible fact switches from B to A. Bars are 95% skeleton-bootstrap intervals.

Model	Not understood Base	Not understood Filter	Not audible Base	Not audible Filter	Transl. (pp)
Gemma-4-31B	45.5	0.1	50.3	0.0	6.3
Qwen3.8-27B	49.6	0.1	57.7	0.5	5.1
GLM-4.7-Flash	46.2	58.5	68.3	71.4	16.3
Nemotron-3-Nano	70.2	35.2	86.4	34.7	-26.0
Mistral-Small-24B	65.8	45.2	72.7	53.0	-0.7
Gemma-4-12B	67.1	0.0	69.0	0.1	-4.9
Phi-4	69.9	10.0	72.7	42.0	-1.0
Falcon-H1-34B	79.1	4.3	74.7	31.4	-5.5
gpt-oss-20b	76.5	6.1	79.3	22.3	0.1
Seed-OSS-36B	73.3	0.1	80.9	0.3	1.4
Qwen3.5-9B	83.9	50.9	85.9	83.2	-12.8

Table 4: Study 1, the list-first prompt (Filter) by model (content and planning questions). Leakage is net use in pp, and the last column is the change in target use when a working translator makes the content accessible.

D Format Experiments

The ablation starts from Study 1's flag-only prompts and adds Study 2's channel field, its direct question and a minimal context in a full factorial design (four models, 96 skeletons, content question, with matched accessible controls). The bridge starts from Study 2's hearing prompts (96 scenes) and swaps in Study 1's system prompt, character sheet, state layout or question one at a time and then together. The replication applies the same swaps to the 96 confirmation scenes. Their characters were invented for Study 2, so each is renamed to Kerry, Rogue or Panam and given the discovery set's short task card, which makes the base prompt match the discovery format on new scenes.

E Human Audit and AI Screening

The 150 replies were drawn with a fixed seed. In each of eight conditions, five from Study 1 and three from Study 2, half came from replies the scorer counted as using the target and half from the rest. The annotator saw only the reply and two candidate facts in random order and marked whether the reply mentions each fact's object or place. Table 29 gives the results by condition. The four disagreements are borderline references such as "the lot" for a parking lot.

Two AI annotators also screened 568 and 624 Study 2 replies. They agreed with the scorer on target use (F1 about 98%) and flagged 11 distinct replies as foil uses that the scorer had missed. Each of these replies names the target values, so the disagreements come from the annotators attaching values to the wrong candidate.

Model	First-step list names target (%)		Translator use change (pp) [95% CI]
	Not understood	Not audible	
Gemma-4-31B	0.0	0.0	6.3 [3.4, 9.6]
Qwen3.8-27B	0.7	13.2	5.1 [-0.7, 11.4]
GLM-4.7-Flash	90.6	90.2	16.3 [11.8, 20.9]
Nemotron-3-Nano	31.1	39.1	-26.0 [-32.1, -19.8]
Mistral-Small-24B	15.3	64.1	-0.7 [-4.8, 3.6]
Gemma-4-12B	0.0	0.2	-4.9 [-10.7, 0.8]
Phi-4	8.6	49.5	-1.0 [-6.4, 4.3]
Falcon-H1-34B	5.8	60.5	-5.5 [-9.5, -1.6]
gpt-oss-20b	2.4	26.5	0.1 [-3.8, 3.9]
Seed-OSS-36B	0.0	0.0	1.4 [-2.6, 5.2]
Qwen3.5-9B	53.4	93.2	-12.8 [-18.6, -7.2]

Table 5: Study 1: target use in the first-step perception list of the list-first prompt, which predicts where the filter fails.

Model	Main prompt	Boundary-instruction	Minimal	Community
Gemma-4-31B	45.5 [38.9, 52.2]	23.1 [18.3, 28.4]	50.6 [44.3, 57.2]	51.0 [44.7, 57.4]
Qwen3.8-27B	49.6 [43.7, 54.9]	23.0 [18.2, 27.6]	46.9 [41.5, 52.0]	51.8 [46.9, 56.4]
Qwen3.5-9B	83.9 [80.5, 87.0]	84.7 [81.4, 87.8]	93.1 [90.7, 95.1]	95.6 [94.3, 96.9]

Table 6: Study 1: language-boundary leakage (repertoire only, pp) under three alternative system prompts. The boundary-instruction prompt tells the model to preserve visible knowledge boundaries.

Subset	Reduction (%)	Transl. (pp)	Holm p
Chinese	66.6	-5.1	0.523
Japanese	66.0	-2.8	0.055
Spanish	61.5	2.0	0.004
Without Kerry	62.9	-0.5	0.004
Without Panam	64.1	-0.5	0.004
Without Rogue	67.0	-4.9	0.464
Two-way bootstrap	64.7	-2.0	0.130

Table 7: Study 1: the list-first test by language, leaving one character out, and with a two-way (skeleton $\times$ character) bootstrap. The joint criterion fails in several subsets.

Variant	Not underst.	Not audible	Reduction (%)	Translator (pp)	Supported
Unconditional (no capability filter)	66.5	72.2	64.7	-2.2	8/8
Soft hedge credit	66.7	72.2	64.7	-1.9	8/8
Exclude truncated	66.7	72.2	64.7	-2.0	8/8
Exclude language/echo flags	67.0	72.2	64.8	-2.3	8/8
Complete responses only	66.7	72.2	64.7	-1.9	8/8
Worst-case failed list-first reply	66.7	72.2	64.6	-2.0	8/8
Both canaries required	47.8	54.5	61.9	1.3	8/8

Table 8: Study 1 scoring and conditioning sensitivities.

Exploratory contrast	Estimate (pp)	95% CI	BH q
Denial while using the content, prose sentence	4.9	[4.3, 5.5]	0.002
Unattributed use, prose sentence minus no language state	7.6	[5.6, 9.6]	0.002
Offline − absent translator	0.1	[-0.5, 0.7]	0.767
Repertoire minus narrated inaudibility (unmatched)	20.3	[19.1, 21.5]	0.002
Few-shot reduction (4 models)	40.4	[37.6, 43.3]	0.002
Direct address − overheard	8.3	[4.9, 11.6]	0.002
Spoken-only − full state	0.2	[-1.2, 1.4]	0.804
Chinese minus Japanese	1.1	[-0.3, 2.7]	0.160
Chinese minus Spanish	-4.5	[-6.9, -2.2]	0.002
Japanese minus Spanish	-5.7	[-8.2, -3.1]	0.002
Explicit admission, prose sentence minus no language state	-0.3	[-0.4, -0.2]	0.002
Admission or attribution, prose sentence minus no language state	-9.4	[-10.8, -8.0]	0.002
Canon speakers, repertoire minus no language state	1.4	[-1.5, 4.6]	0.392
Weak canon non-speakers, repertoire minus no language state	-3.4	[-6.6, 0.3]	0.109
Non-speakers with a name prior, repertoire minus no language state	-4.5	[-8.9, -0.5]	0.045
Present-day world − main set	-16.4	[-24.9, -7.8]	0.002

Table 9: Study 1 exploratory family (Benjamini–Hochberg, $q = .10$). No language state means the state omits the character's languages. The world comparison uses different scenario sets.

Family / component	Estimate (pp)	95% CI	Null	Holm p
Channel, Hear	72.27	[70.90, 73.57]	0	0.0015
Channel, Understand	47.20	[44.86, 49.61]	0	—
Channel, See	69.27	[67.25, 71.35]	0	—
Channel, Receive	67.58	[65.82, 69.34]	0	—
State-correct net use	63.95	[62.73, 65.19]	0	0.0015
View, full minus view leakage	64.05	[62.81, 65.27]	0	0.0015
View minus full accessible use	0.49	[-0.18, 1.19]	-5	—
View minus hard event retention	67.75	[65.36, 70.05]	0	—

Table 10: Study 2 confirmation (four models, 96 original-character scenes). The channel test requires all four channels to be positive, the state-correct test uses replies whose state probe is correct (6,068), and the view test requires the view to reduce leakage, keep accessible use within 5 pp, and keep more observations than removal. One Holm-adjusted p per family.

Model	Hear	Understand	See	Receive
Gemma-4-31B*	5.5	0.0	16.4	22.4
Qwen3.8-27B	22.4	2.6	41.4	14.8
Mistral-Small-24B	95.6	1.6	31.0	83.3
Gemma-4-12B	99.5	14.1	24.0	78.6
Seed-OSS-36B	59.6	58.6	53.1	60.4
GLM-4.7-Flash*	87.2	71.1	66.4	52.6
Falcon-H1-34B	64.1	74.7	54.9	97.9
Phi-4	97.1	14.1	93.5	98.2
gpt-oss-20b	94.8	37.2	91.4	88.8
Qwen3.5-9B*	97.4	51.3	96.6	96.4
Nemotron-3-Nano*	99.0	66.4	97.7	99.0
All 11 models	74.7	35.6	60.6	72.0
7 added models	76.2	29.0	55.6	74.6

Table 11: Study 2 confirmation set (96 original-character scenes), net use of inaccessible content under full context (pp). *Locked four-model cohort.

Model	Hear	Understand	See	Receive	State correct	Conditional D
Gemma-4-31B	5.47	0.00	16.41	22.40	100.00	11.07
GLM-4.7-Flash	87.24	71.09	66.41	52.60	99.87	69.30
Nemotron-3-Nano-30B	98.96	66.41	97.66	98.96	95.18	90.01
Qwen3.5-9B	97.40	51.30	96.61	96.35	100.00	85.42

Table 12: Confirmation model detail. Boundary columns and conditional D are net canary-use points, and state correctness is a percentage. Every model has eligible role endpoints.

Model	Channel	No correct (/192)	Yes correct (/192)	False Yes	False No
Gemma-4-31B	Hear	192 (100.00%)	192 (100.00%)	0	0
Gemma-4-31B	Understand	192 (100.00%)	192 (100.00%)	0	0
Gemma-4-31B	See	192 (100.00%)	187 (97.40%)	0	5
Gemma-4-31B	Receive	192 (100.00%)	192 (100.00%)	0	0
GLM-4.7-Flash	Hear	191 (99.48%)	192 (100.00%)	1	0
GLM-4.7-Flash	Understand	192 (100.00%)	192 (100.00%)	0	0
GLM-4.7-Flash	See	192 (100.00%)	178 (92.71%)	0	14
GLM-4.7-Flash	Receive	192 (100.00%)	190 (98.96%)	0	2
Nemotron-3-Nano-30B	Hear	155 (80.73%)	192 (100.00%)	37	0
Nemotron-3-Nano-30B	Understand	192 (100.00%)	192 (100.00%)	0	0
Nemotron-3-Nano-30B	See	192 (100.00%)	141 (73.44%)	0	51
Nemotron-3-Nano-30B	Receive	192 (100.00%)	192 (100.00%)	0	0
Qwen3.5-9B	Hear	192 (100.00%)	192 (100.00%)	0	0
Qwen3.5-9B	Understand	192 (100.00%)	192 (100.00%)	0	0
Qwen3.5-9B	See	192 (100.00%)	175 (91.15%)	0	17
Qwen3.5-9B	Receive	192 (100.00%)	192 (100.00%)	0	0

Table 13: Independent confirmation access probes in both states. No is the correct inaccessible answer and Yes the correct accessible answer. Each cell contains 96 scenes times two versions, and all 6,144 outputs are valid. False Yes and False No count the two error directions. Model-level accuracy averages four equally sized channels; always No gives 50% on the balanced 3,072/3,072 population.

Boundary	State-correct D [95% CI]	N	Capability-eligible D [95% CI]	N
Hear	72.19 [70.79, 73.50]	1460	72.19 [70.79, 73.50]	1460
Understand	47.20 [44.86, 49.61]	1536	50.98 [48.36, 53.67]	1152
See	69.27 [67.25, 71.35]	1536	69.27 [67.25, 71.35]	1536
Receive	67.58 [65.82, 69.34]	1536	67.58 [65.82, 69.34]	1536

Table 14: State-correct boundary decomposition in the four-model confirmation. Values are pp. The .85 gate applies only to comprehension, where all state probes are correct. Eligibility is fixed from translation results, and model-specific ratios are recomputed over the original 96-scene bootstrap.

Model	Chinese	Japanese	Spanish
Gemma-4-31B	100.00	100.00	100.00
GLM-4.7-Flash	100.00	62.50	100.00
Nemotron-3-Nano-30B	100.00	75.00	87.50
Qwen3.5-9B	100.00	62.50	100.00

Table 15: Both-canary translation use on 64 original-character prompts per model–language unit (%). The diagnostic threshold is 85%, and the primary analysis is unconditional.

Boundary	Membership minus flag	Prose+flag minus flag
Hear	17.52 [14.96, 20.08]	-37.59 [-41.29, -33.90]
Understand	28.60 [25.47, 31.82]	-3.98 [-8.05, -0.09]
See	32.86 [28.98, 36.84]	-17.52 [-22.54, -13.35]
Receive	3.41 [0.57, 6.34]	-37.88 [-41.38, -34.09]

Table 16: Representation effects on inaccessible net use (pp), with 95% intervals. Eleven completed discovery models, 24 matched scenes. The same comprehension prose contrast is +5.73 [0.00, 11.98] in the four-core subset.

Boundary	Method	Inaccessible D	Accessible target use
Hear	Full	57.20 [54.69, 59.55]	89.67 [86.63, 92.10]
Hear	Prose+flag	34.38 [31.16, 37.59]	93.40 [90.71, 95.66]
Hear	Self-filter	47.74 [44.62, 50.87]	93.49 [91.06, 95.66]
Hear	Hard removal	0.09 [0.00, 0.26]	89.06 [85.42, 92.01]
Hear	Character view	0.09 [0.00, 0.26]	88.98 [85.33, 92.01]
Understand	Full	30.99 [27.69, 33.94]	91.49 [88.54, 94.01]
Understand	Prose+flag	32.29 [29.34, 35.07]	95.57 [93.66, 97.31]
Understand	Self-filter	2.00 [0.87, 3.39]	62.93 [59.64, 66.06]
Understand	Hard removal	0.00 [0.00, 0.00]	87.76 [84.11, 90.97]
Understand	Character view	-0.09 [-0.26, 0.00]	87.76 [84.20, 90.97]
See	Full	45.66 [41.49, 49.48]	79.34 [75.17, 83.16]
See	Prose+flag	27.17 [24.57, 29.60]	96.61 [95.23, 97.74]
See	Self-filter	22.22 [19.79, 24.57]	81.51 [78.21, 84.46]
See	Hard removal	0.00 [0.00, 0.00]	80.56 [76.39, 84.38]
See	Character view	-0.09 [-0.26, 0.00]	80.47 [76.39, 84.38]
Receive	Full	56.77 [53.65, 59.64]	94.18 [90.97, 96.61]
Receive	Prose+flag	31.25 [29.17, 33.42]	96.79 [95.57, 97.83]
Receive	Self-filter	4.77 [3.21, 6.60]	53.04 [48.87, 57.21]
Receive	Hard removal	-0.09 [-0.26, 0.00]	90.89 [87.59, 93.66]
Receive	Character view	0.00 [-0.26, 0.26]	90.89 [87.59, 93.66]

Table 17: Five methods in the complete prespecified six-model discovery cohort on 48 matched scenes (1,152 endpoints per cell). Leakage is net pp and accessible use is a percentage. Intervals are 95% paired-bootstrap intervals.

Method	Leakage	Access. use	Event retention
Full context	64.08	92.82	73.26
Hard removal	0.03	93.33	32.25
Character view	0.03	93.31	100.00

Table 18: Confirmation cohort. Leakage is net D in pp, and accessible target use and partial-event retention are percentages. Retention excludes Hear. Full-context and projected inputs may have different schemas even in accessible controls.

Boundary	Method	Delay 0	Delay 2	Delay 4
Hear	full	69.14 [65.23, 72.66]	68.75 [65.62, 71.88]	71.09 [67.97, 73.83]
Hear	view	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Understand	full	42.58 [36.72, 48.05]	52.73 [48.05, 57.03]	58.20 [53.12, 63.28]
Understand	view	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
See	full	55.47 [49.61, 61.33]	50.00 [44.14, 55.86]	50.78 [43.75, 57.03]
See	view	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Receive	full	67.97 [63.67, 71.88]	68.36 [63.67, 72.66]	69.53 [66.02, 72.66]
Receive	view	-0.39 [-1.17, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]

Table 19: Delayed net use (pp), 32-scene subset, with 95% intervals. The original event remains in context throughout the scripted exchanges.

Boundary	Method	Fixed-A choice contrast	95% CI	Valid choices (%)
Hear	full	63.28	[53.91, 71.09]	99.61
Hear	view	0.78	[0.00, 2.34]	96.88
Understand	full	54.69	[47.66, 60.94]	100.00
Understand	view	0.00	[0.00, 0.00]	100.00
See	full	49.22	[42.97, 54.69]	100.00
See	view	0.00	[0.00, 0.00]	99.22
Receive	full	55.47	[46.88, 64.06]	99.61
Receive	view	0.00	[0.00, 0.00]	97.66

Table 20: Fixed-A decision contrast (pp) and response validity. The choice of location A is tracked across versions, independently of its option number.

Boundary	Method	A first (18 scenes)	A second (14 scenes)	Equal-position effect
Hear	full	65.28 [56.94, 72.22]	60.71 [42.86, 75.00]	63.00 [53.27, 71.23]
Hear	view	0.00 [0.00, 0.00]	1.79 [0.00, 5.36]	0.89 [0.00, 2.68]
Understand	full	56.94 [50.00, 63.89]	51.79 [39.29, 62.50]	54.37 [47.12, 60.91]
Understand	view	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
See	full	50.00 [43.06, 56.94]	48.21 [39.29, 57.14]	49.11 [42.76, 54.86]
See	view	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Receive	full	59.72 [50.00, 69.44]	50.00 [35.71, 62.50]	54.86 [45.53, 63.10]
Receive	view	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]

Table 21: Option-position sensitivity (pp, 95% intervals). A/B order and the question are identical within every pair. The equal-position estimate weights the two order strata equally and resamples scenes independently within each stratum. Original paired effects and tests remain unchanged.

Input component	Exact saved text or specified substitution
Role system prefix	`You are role-playing Mira Sen. Use the character sheet and the scene to reply to your companion in one short English spoken turn. Return only the spoken words.`
Shared character	`[Character] Mira Sen is an original fictional volunteer in Harrowbay, a contemporary coastal civic district. They are practical, polite, and concise. They rely on their own observations. Their history and relations are defined only by this scene.`
Full scene (A, no access)	`[Scene state] {"character_id": "Mira Sen", "events": [{"actor": "Alex", "audible_to_Mira Sen": false, "boundary_channel": "Hear", "content": "The glove is in the church.", "event_id": "e1", "kind": "speech", "language": "English"}], "observable_event_shell": null, "scene": "An original community team is preparing an activity. Its members have not shared the latest finding."}`
Role user message	`What information did you get about the item and its location?`
Independent probe prefix	`Read the scene as an evaluator. Answer the access question with only Yes or No.`
Independent probe user	`Did Mira Sen hear the utterance?`
Counterfactual B	Replace `The glove is in the church.` with `The broom is in the stairwell.`
Accessible control	Replace `audible_to_Mira Sen`: `false` with `true`.
Inaccessible view scene	`[Scene state] {"character_id": "Mira Sen", "observations": [], "scene": "An original community team is preparing an activity. Its members have not shared the latest finding."}`

Table 22: Exact messages for the selected hearing case, with A/B and access-state substitutions. The probe shares the full scene and character but has its own prefix and question. View shares the role prefix and question and substitutes the displayed observation JSON. Accessible view contains the permitted event in its observation list.

Version	Access	Probe	Full-context spoken reply	Character-view spoken reply
A	No	No	I heard Alex mention the glove is in the church, though he wasn't speaking to me directly.	I haven't received the latest findings yet; the team members haven't shared them.
B	No	No	I heard Alex mention the broom is in the stairwell, though they weren't speaking to me directly.	I haven't received the latest findings yet; the team members haven't shared them.
A	Yes	Yes	Alex mentioned the glove is in the church.	Alex mentioned the glove is in the church.
B	Yes	Yes	Alex mentioned the broom is in the stairwell.	Alex just told me the broom is in the stairwell.

Table 23: All saved content replies and independent probe answers for scene C001, Qwen3.5-9B, access flag, no delay. Full replies score $(T, F, D) = (1, 0, 1)$ in all four rows, inaccessible view scores $(0, 0, 0)$ and accessible view $(1, 0, 1)$.

Model	Base	Sys.	Sheet	State	Quest.	All	Audible use
Gemma-4-31B	0	0	1	**61**	**14**	**100**	83–100
Qwen3.8-27B	9	10	**68**	**19**	9	**78**	80–100
Falcon-H1-34B	31	26	**70**	**87**	**100**	**100**	98–100
Seed-OSS-36B	60	**47**	**85**	**71**	**100**	**95**	95–100
Mistral-Small-24B	72	**64**	73	73	**100**	**100**	77–100
Gemma-4-12B	78	**88**	**86**	76	**100**	**100**	77–100
gpt-oss-20b	80	85	**92**	82	72	**99**	92–99
Qwen3.5-9B	85	80	**98**	90	**100**	**99**	95–100
GLM-4.7-Flash	88	86	**93**	**68**	**100**	**98**	69–100
Nemotron-3-Nano	97	**99**	97	**100**	**100**	99	98–100
Phi-4	98	**95**	**94**	**86**	100	100	93–100

Table 24: Bridge experiment on the discovery scenes (96 scenes, 384 prompts per swap). Columns give net use of unheard speech (pp) under Study 2's prompt (Base), with one Study 1 feature swapped in, and with all four. Bold marks a change from Base whose paired 95% skeleton-bootstrap interval excludes zero. The last column is the range of target use for the same speech when audible (%).

Model	Base	Sys.	Sheet	State	Quest.	All	Audible use
Gemma-4-31B	0	0	0	**96**	**8**	**100**	92–100
Qwen3.8-27B	6	10	**72**	**12**	**0**	**71**	96–100
Falcon-H1-34B	25	21	**78**	**98**	**98**	**100**	100
Seed-OSS-36B	57	54	**74**	**73**	**98**	**90**	88–100
Mistral-Small-24B	81	79	79	**90**	**100**	**100**	98–100
GLM-4.7-Flash	97	99	96	**91**	**100**	97	95–100
Phi-4	98	**100**	96	**100**	**100**	**100**	96–100
Gemma-4-12B	100	100	100	100	100	100	100
gpt-oss-20b	100	**89**	**90**	99	**87**	100	94–100
Nemotron-3-Nano	100	100	**92**	100	100	100	92–100
Qwen3.5-9B	100	100	**98**	100	100	100	100

Table 25: Bridge experiment on the confirmation scenes, with each character renamed to a Study 1 persona (96 scenes, 384 prompts per swap). Columns give net use of unheard speech (pp) under Study 2's prompt (Base), with one Study 1 feature swapped in, and with all four. Bold marks a change from Base whose paired 95% skeleton-bootstrap interval excludes zero. The last column is the range of target use for the same speech when audible (%).

Model	Base	Sys.	Sheet	State	Quest.	All
Claude Sonnet 5.5	0.5	**9.9**	**15.1**	**8.9**	**9.4**	**13.0**
Gemini 3.1 Pro	0.0	–	0.0	0.0	–	**21.4**

Table 26: Single-feature swaps for two hosted models on the discovery hearing prompts (192 unheard prompts per cell, net use in pp). Bold marks a change from Base whose paired 95% interval excludes zero. Gemini 3.1 Pro was run on the two swaps that moved the open models most.

Model	Repertoire flag	Audibility flag	Compr. flag	Prose sentence	Control use	State probe	Transl. probe	Bridge base → all
GPT-6 Sol	0.2	0.0	0.9	0.2	100.0	100.0	97.9	0.0 → 0.0
Claude Opus 5.5	2.1	2.1	–	–	99.3	100.0	92.7	0.0 → 1.0
Claude Sonnet 5.5	26.0	8.0	35.8	21.7	99.0	96.5	95.1	0.5 → 13.0
Gemini 3.1 Pro	1.6	9.0	–	–	99.7	100.0	97.2	0.0 → 21.4
Gemini 3.8 Flash	0.2	-0.2	0.0	0.0	99.0	100.0	93.8	0.0 → 0.0
Grok 4.7	1.0	0.0	1.0	0.3	100.0	100.0	99.3	0.0 → 0.0
DeepSeek-V4-Pro	71.2	78.3	43.4	21.2	93.8	100.0	96.2	6.2 → 67.2

Table 27: Hosted models and DeepSeek-V4-Pro on Study 1 core conditions (content question, 288 items per cell) and on both ends of the bridge. Leakage is net use in pp, control use and probes are percentages. Claude Opus 5.5 and Gemini 3.1 Pro were not run on the comprehension-flag and prose conditions.

Model	Study 1 prompt	+ channel	+ question	+ minimal	All three
Flag: not audible					
Gemma-4-31B	52.1	52.1	51.0	54.7	50.0
Qwen3.8-27B	71.4	74.0	80.2	51.6	32.8
Seed-OSS-36B	90.1	88.0	100.0	87.0	83.3
Phi-4	99.0	100.0	94.3	100.0	94.3
Flag: not understood					
Gemma-4-31B	6.2	3.1	1.6	3.1	0.0
Qwen3.8-27B	35.9	34.4	39.6	12.0	14.1
Seed-OSS-36B	69.8	67.2	95.3	37.0	31.8
Phi-4	88.5	89.6	66.1	69.8	49.5

Table 28: Format ablation from Study 1 toward Study 2 (96 skeletons, content question): net use (pp) when one Study 2 feature is added (a field naming the channel, the direct question, a minimal context) and with all three.

Condition	n	Human D	Scorer D	Precision	Recall
Study 1, list-first answer	14	25.1	25.1	100.0	100.0
Study 1, control, fluent	14	75.5	75.5	100.0	100.0
Study 1, language, prose sentence	10	33.9	33.9	100.0	100.0
Study 1, language, repertoire only	30	66.7	66.7	100.0	100.0
Study 1, not audible, flag only	20	75.0	72.2	100.0	96.3
Study 2, character view, inaccessible	12	0.4	0.5	83.3	100.0
Study 2, full context, accessible	10	97.1	95.1	100.0	98.0
Study 2, full context, inaccessible	40	60.9	60.9	100.0	100.0

Table 29: Blind human audit (150 replies, one annotator). D is net use (pp) reweighted to each condition. Precision and recall are the scorer's for target use against the human labels, weighted the same way. Neither the annotator nor the scorer found a foil use.